INCH-POUND
MIL-DTL-32629
8 May 2019

DETAIL SPECIFICATION

CLOTH, BANNER, NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This document covers one (1) type of nylon banner cloth.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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THE INSTITUTE OF HERALDRY, DEPARTMENT OF THE ARMY

Department of Defense Standard Shades for Nylon Flag Bunting

(Copies of this publication can be obtained by request through The Institute of Heraldry, 9325 Gunston Road, Room S-112, Fort Belvoir, VA 22060; contact by email: usarmy.beloir.hqda.mbx.tioh-webmaster@mail.mil or by telephone: 571-515-0346.)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov/)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 5, Fabric Hand: Guidelines for Subjective Evaluation of
AATCC Evaluation Procedure 8, AATCC 9 Step Chromatic Transference Scale Rating
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 16.3 - Colorfastness to Light: Xenon
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 81 - pH of the Water-Extract from Bleached Textiles
AATCC Test Method 107 - Colorfastness to Water
AATCC Test Method 132 - Colorfastness to Drycleaning
AATCC Test method 135 - Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available on line at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method

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ASTM D2261	- Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)
ASTM D3774	- Standard Test Method for Width of Textile
ASTM D3775	- Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3990	- Standard Terminology Relating to Fabric Defects
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5430	- Standard Test Methods for Visually Inspecting and Grading Fabrics

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and 6.3.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Nylon. The cloth shall be constructed with nylon, multifilament, 70 denier warp x 200 denier filling and meet the requirements of Table II when tested as specified in Table IV.

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3.4.1.1 Yarn. The yarn shall be 1-ply for both the warp and filling.

3.4.1.2 Weave. The cloth shall be a 2 x 2 twill.

3.5 Color. The color shall be as specified in the contract in 6.2 and shall match the standard sample (see 6.4). Shade designations and PMS numbers are from the DoD Standard Shades for Nylon Flag Bunting as obtained from The Institute of Heraldry Department of the Army.

3.5.1 Visual shade matching. The color and appearance of the cloth specified in 3.4 shall match the standard sample when tested as specified in 4.5.1.

3.5.2 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished cloth referenced in 3.4 shall be equal to or better than the standard sample and conform to the colorfastness requirements listed below in Table I when tested as specified in Table IV.

TABLE I. Colorfastness requirements.

Colors Evaluation	Laundering (after 1 cycle) Color Change and Staining (min.)	Light (after 40 AFU or 170 kJ/ (m^2nm)@420 nm) (min.) 1/	Crocking Dry/Wet (min.)	Wet Dry cleaning (1 cycle) (min.)	Water Staining (min.)
All colors	3-4	3-4	3.5/3.5	3-4	3-4

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished cloth shall conform to the requirements specified in Table II when tested as specified in 4.5.

TABLE II. Physical requirements.

Characteristic	Requirement
Weight (oz./sq.yd.)	3.7 - 3.9
Yarns per inch (min.)	
Warp	164
Filling	82
Breaking Strength, lbs. (min.)	
Warp	215
Filling	254
Tongue Tear, lbs. (min.)	
Warp	4.5
Filling	6.0
Air Permeability (cu.ft./min./sq.ft.)	7.0 - 10. 0

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3.6.1 Width (for Government purposes only), Unless otherwise specified in the contract, the width of the cloth shall be a minimum of 60 inches (see 6.2) inclusive of selvages when fly shuttle looms are used, and exclusive of selvages and fringe when shuttleless looms are used. Tuck-in selvages shall not be permitted.

3.7 Finish. The cloth shall have a hand and drape equal to or better than the standard sample when evaluated as specified in 4.4.4.

3.7.1 pH. The pH value of the water extract of the finished cloth shall be between 5.5 and 8.5, inclusive, when tested as specified in 4.5.

3.8 Length and put-up (for Government purposes only). Unless otherwise specified (see 6.2), the cloth shall be furnished full width in continuous lengths, each not less than 50 yards.

3.9 Face Identifcation. The warp side shall be identified as the face side by stamping that side with the word "FACE" at each end of the roll.

3.10 Fiber identification. Each roll of finished cloth shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the with the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the acceptance quality levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects and tested for the characteristics as specified in Table II.

4.3 Conformance inspection. Conformance inspection shall include the examinations in 4.4.1 through 4.4.3 and the testing in Table II, as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and the acceptance quality limits as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification, or applicable procurement documents, all inspections shall be performed in accordance with all the requirements of referenced documents,

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4.4 Inspection methods.

4.4.1 Yard by yard examination. Each roll in the sample shall be examined yard-by-yard (full width excluding selvage) on both sides with both overhead and back lighting. The defects are as defined in ASTM D3990 and only defects that are clearly noticeable at normal inspection distance (3-feet) shall be scored. The defects are assigned demerit points in accordance with ASTM D5430, Option A, with the exceptions listed as follows:

a. The defects found shall be counted regardless of their proximity to each other, except where two (2) or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted.

b. Only coarse yarns that exceed twice the normal diameter shall be scored.

c. Mixed filling shall be scored only if attributed to shade bar resulting from wrong ply, wrong twist in the yarn or off shade yarn.

d. The defects specified in Table III shall be scored four (4) points for each yard in which they occur.

TABLE III. Defects that score four (4) points.

Defect Description
Any hole, cut, tear or abrasion mark
Edge ravels when pulled outward
Uneven weaving throughout
Broken or missing yarn
Any spot, stain or streak more than 1-inch in combined directions
Knots great than Fabric Defect Replica Scale Level D (see 2.3)
Slubs greater than Fabric Defect Replica Scale Level C (see 2.3)
Area of poor dye penetration, streak or shade bar
Width not as specified with a tolerance of no less than 1/2-inch
Objectionable odor
Slack or tight selvages
Any embedded foreign matter or uncleanness
Edges rolled, folded, scalloped or corded, baggy, ridgy or wavy cloth

4.4.2 Shade and finish appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll in the sample off shade; shaded side to side, side to center, or end to end; or not having the same finished appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.4.2.1 Shade match and appearance examination. The shade match and appearance examination shall be as specified in the contract or purchase order.

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4.4.3 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects. The lot shall be unacceptable if it is not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act, or the face stamping is missing, on the wrong side, incorrect or illegible.

4.4.4 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing specified, wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.5 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a length defect. The lot shall be rejected if two (2) or more rolls in the sample are defective with respect to length or if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table IV and the methods of testing specified, wherever applicable and as listed in Table IV shall be followed.

TABLE IV. End item testing.

Characteristic	Requirement reference	Test method
Fiber Identification	3.4.1	AATCC 20 1/
Multifilament	3.4.1	Visual 2/
Denier (Warp and Filling)	3.4.1	ASTM D1907/D1907M
Yarn Ply (Warp and Filling)	3.4.1.1	Visual 2/
Weave	3.4.1.2	Visual 2/
Visual Shade Matching	3.5.1	4.3.1

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TABLE IV. End item testing. - Continued

Characteristic	Requirement reference	Test method
Colorfastness to:	3.5.2	
Laundering (1 cycle)	Table I	AATCC 61 (1A) 3/
Light (40 AFU or 170 kJ/(m^2nm) @420 nm)	Table I	AATCC 16.3, Option 3 4/
Crocking (Wet & Dry)	Table I	AATCC 8 5/
Wet Dry Cleaning (1 cycle)	Table I	AATCC 132 3/
Water	Table I	AATCC 107 3/
Weight	3.6 & Table II	ASTM D3776/D3776M
Yarns per inch (Warp and Filling)	3.6 & Table II	ASTM D3775
Breaking Strength (Warp and Filling)	3.6 & Table II	ASTM D5034
Tongue Tear (Warp and Filling)	3.6 & Table II	ASTM D2261
Air Permeability (cu.ft./min./sq.ft.)	3.6 & Table II	ASTM D737
Width	3.6.1	ASTM D3774
Finish	3.7	AATCC Eval. Proc. 5 6/
pH	3.7.1	AATCC-81

1/ Unless otherwise specified, a certificate or compliance shall be submitted and will be acceptable for the stated requirement (see 6.6).
2/ One (1) determination for each sample unit and the result reported as "pass" or "fail".
3/ Rated using AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
4/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change
5/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale.
6/ The hand and drape of the sample shall be compared to the standard sample.

4.5.1 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The nylon banner cloth is intended for use in the manufacture of various types of flags.

6.2 Acquisition requirements. Acquisition documents should specify the following
a. Title, number, and date of this specification.
b. Color required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.2, 6.3)
e. Width required, if other than specified (see 3.6.1)
f. Minimum length, if other than specified (see 3.8)
g. Conformance inspection conditions (see 4.3)
h. Inspection conditions (see 4.4)
i. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Replica set. The SDL Atlas Slub/Knot Replica Set is not available from their website at this time. Recommend calling customer service, 1-803-329-2110.

6.6 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Subject term (key word) listing.

Bunting
Flags

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Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:
Navy - MC
DLA - CT

Civil Agencies
GSA

Preparing activity:

DLA - CT

(Project: 8305-2019-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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